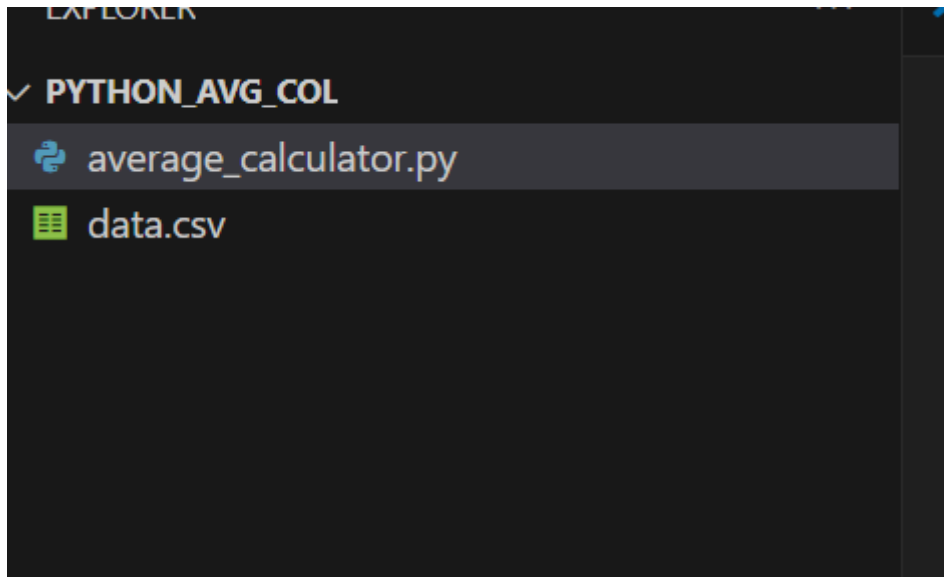


//Q2

//<https://github.com/AkashChand6n/Devops-retest.git>

Python script that reads a CSV file and calculates the **average value** of a **specified column**



```
Welcome | average_calculator.py X | data.csv
average_calculator.py
1  # Author: Akash Chandran
2  # Description: Read a CSV file and calculate the average of a specified column.
3
4  import csv # Import the CSV module to read CSV files
5
6  # Define a function to calculate average of a specific column in a CSV file
7  def calculate_average(filename, column_name):
8      try:
9          # Open the CSV file for reading
10         with open(filename, 'r') as csvfile:
11             reader = csv.DictReader(csvfile) # Read file into a dictionary per row
12             values = []
13
14             for row in reader:
15                 # Check if the specified column exists in the current row
16                 if column_name in row:
17                     try:
18                         # Try converting the value to float and store it
19                         value = float(row[column_name])
20                         values.append(value)
21                     except ValueError:
22                         # If conversion fails, print warning and skip the value
23                         print(f"Warning: Non-numeric value skipped in row: {row}")
24                 else:
25                     # If column not found in row, show error and stop
26                     print(f"Column '{column_name}' not found in row: {row}")
27             return
28
29     # Check if we have collected any valid values
```

```

29         # Check if we have collected any valid values
30         if not values:
31             print(f"No valid numeric data found in column '{column_name}'.")
32             return
33
34         # Calculate and print the average
35         average = sum(values) / len(values)
36         print(f"Average of column '{column_name}': {average}")
37
38     except FileNotFoundError:
39         # If the file doesn't exist
40         print(f"Error: File '{filename}' not found.")
41     except Exception as e:
42         # Handle any other unexpected errors
43         print(f"An unexpected error occurred: {e}")
44
45 # Entry point of the script
46 if __name__ == "__main__":
47     # Call the function with the CSV filename and column name
48     calculate_average('data.csv', 'score')
49

```

Author: Akash Chandran

Description: Read a CSV file and calculate the average of a specified column.

```
import csv # Import the CSV module to read CSV files
```

```
# Define a function to calculate average of a specific column in a CSV file
```

```
def calculate_average(filename, column_name):
```

```
    try:
```

```
        # Open the CSV file for reading
```

```
        with open(filename, 'r') as csvfile:
```

```
            reader = csv.DictReader(csvfile) # Read file into a dictionary per row
```

```
            values = []
```

```
            for row in reader:
```

```
                # Check if the specified column exists in the current row
```

```
                if column_name in row:
```

```
                    try:
```

```

        # Try converting the value to float and
store it        value = float(row[column_name])
                values.append(value)
            except ValueError:
                # If conversion fails, print warning and
skip the value    print(f"Warning: Non-numeric value skipped
in row: {row}")
            else:
                # If column not found in row, show error and
stop            print(f"Column '{column_name}' not found in row:
{row}")
                return

        # Check if we have collected any valid values
        if not values:
            print(f"No valid numeric data found in column
'{column_name}'.")
            return

        # Calculate and print the average
        average = sum(values) / len(values)
        print(f"Average of column '{column_name}': {average}")

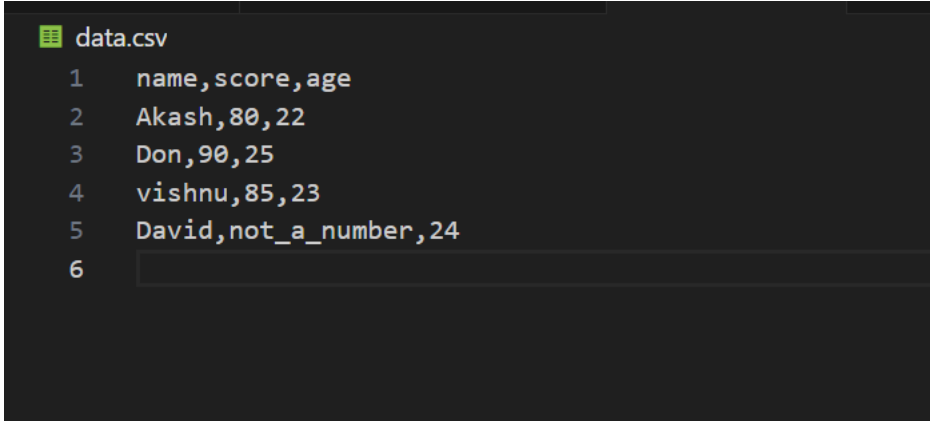
    except FileNotFoundError:
        # If the file doesn't exist
        print(f"Error: File '{filename}' not found.")
    except Exception as e:
        # Handle any other unexpected errors
        print(f"An unexpected error occurred: {e}")

# Entry point of the script
if __name__ == "__main__":

```

```
# Call the function with the CSV filename and column name  
calculate_average('data.csv', 'score')
```

sample data



```
data.csv  
1  name,score,age  
2  Akash,80,22  
3  Don,90,25  
4  vishnu,85,23  
5  David,not_a_number,24  
6
```

```
\Desktop\python_avg_col> python3 average_calculator.py
```

```
Warning: Non-numeric value skipped in row: {'name': 'David', 'score': 'not_a_number', 'age': '24'}  
Average of column 'score': 85.0
```